

5.4 Adding and subtracting algebraic terms

Here is an algebraic expression with three terms: $3a - 5b + 4$

The three terms are $3a$, $-5b$ and 4 .

Terms are always separated by a plus (+) or minus (-) sign.

$3a$ and $7a$ are called **like terms** because they have the **same pronumerals**.

1 Circle the **like terms** in each list.

a $5a$ $8a$ $4b$

b $3b$ $3a$ $7a$

c $3y$ $8x$ $7y$

d $2y$ $3x$ x

e $2m$ $-3m$ $5k$

f $8k$ $-7m$ $3k$

g $-2x$ $-4y$ $6x$

h $5n$ 3 $-8n$

i -5 $4k$ $8k$

j $3x$ $4y$ $5x$ 6

k $2a$ $4b$ -6 $8b$

l -3 $4k$ $-2k$ $7m$

m $4a$ -5 7 $3m$

n $4mn$ $3a$ $7mn$

o $9a$ $2ab$ $-3b$ $11ab$

Only **like terms** can be added or subtracted.

This is called **collecting like terms**.

EXAMPLE

$$3n + 2n$$

$$= 5n$$



$$2n + 4 + n$$

$$= 3n + 4$$



$$2a + n + 3n$$

$$= 2a + 4n$$



2 Collect like terms in these expressions.

a $4a + 3a =$ _____

b $5n + 2n =$ _____

c $8k + 3k =$ _____

d $9m + 4m =$ _____

e $12y + 7y =$ _____

f $9x + x =$ _____

g $13k + 9k =$ _____

h $m + m =$ _____

i $5y + 11y =$ _____

j $7a + 5a =$ _____

k $9g + 6g =$ _____

l $8x + 13x =$ _____

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3 Collect like terms in these expressions.

a $9n - 5n =$ _____

b $7a - 3a =$ _____

c $5m - 3m =$ _____

d $11x - 6x =$ _____

e $9m - 8m =$ _____

f $12y - 8y =$ _____

g $8a - a =$ _____

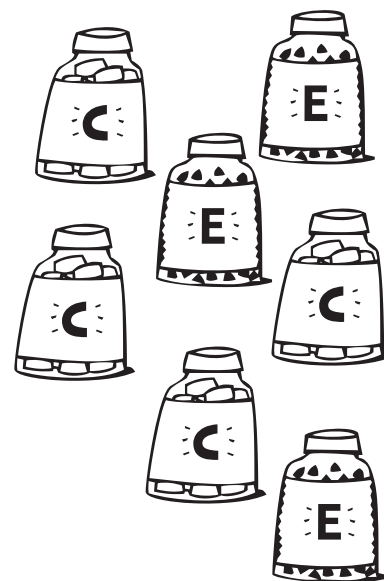
h $13n - 5n =$ _____

i $7a - 7a =$ _____

j $10y - 9y =$ _____

k $16n - 9n =$ _____

l $15m - m =$ _____



4 Collect like terms to simplify these expressions.

a $6m + 5m + 2a =$ _____

b $5n + 2n + 4 =$ _____

c $4n + 9 + 3 =$ _____

d $8x + 2y + 4y =$ _____

e $3a + 6 + 2a =$ _____

f $4a + a + 2a =$ _____

g $11k + 9k + 3m =$ _____

h $6m - 3m + 7m =$ _____

i $5n + 8 - 2n =$ _____

j $5y + 6 - 3 =$ _____

k $16x - 3x - 7x =$ _____

l $12m - 7k - 9m =$ _____

m $11xy + 5xy - 6xy =$ _____

n $7ab - 2ab - ab =$ _____

o $8k + 5k - 5m =$ _____

p $11m - 7a - 4m =$ _____

5 Collect like terms to simplify these expressions.

a $5n + 2 + 3n + 4 =$ _____

b $3a + 5 + 2a + 6 =$ _____

c $8x + 2y + 3x + y =$ _____

d $6a + 5b + a + 2b =$ _____

e $4k + 8m + 6k + 2m =$ _____

f $3x + 7y + 9y + 5x =$ _____

g $8n + 4 - 3n + 6 =$ _____

h $4m + 5 - 2m - 3 =$ _____

i $12a + 5 - 4 - 8a =$ _____

j $6x + 3y - y + 4x =$ _____

k $9k + 2m - 7k - m =$ _____

l $5ab + 11 - 6 - ab =$ _____